PURDUE UNIVERSITY | Aeronautics and Astronautics
COLLEGE OF ENGINEERING

Course Information

Course number and title: **AAE 40000 – Undergraduate Senior Seminar, Spring 2026**

CRN: **67991**

Meeting day and time: **F 12:30pm – 1:20pm**

Meeting location: **Max W & Maileen Brown Hall (BHEE) Room 129**

Instructional Modality: **Face-to-Face (in person)**

Course credit hours: **1.000**

Prerequisites: **senior standing**

Instructor Contact Information

Instructor: **Prof. William Crossley**

Office: **ARMS 2023**

Phone number: **(765) 496-2872**

Email: **crossley@purdue.edu** **(this is the best way to reach me)**

Office hours: Because there are no typical assignments for the seminar course, the course does not have scheduled office hours. However, please do contact me via email, with copy to the School's senior administrative assistant, Audrey Sherwood (asherwo@purdue.edu), if you want to arrange a discussion with me. My availability somewhat is limited because of my duties as Head; I would like to be able to have discussions with everyone, but this may not always be possible. The meetings can be in-person or virtual, depending on my availability and your preferences.

Course Description

The purpose of this course is to help you succeed in your final year at Purdue, provide additional opportunities to broaden your perspectives about aerospace engineering, and give you a head start on planning your future after graduation.

Guest Speakers

Most of the seminar meetings will feature a guest speaker. We will mostly have in-person speakers during the scheduled course time (Fridays 12:30pm to 1:20pm Eastern).

In a large class section like AAE 400, it is difficult to have everyone ask questions verbally. To assist with this, we will also use a Piazza Live Q&A to collect questions from you for the speaker, and I will share these with the speaker at the end of the session. Do not ask or post intentionally insensitive or inflammatory questions on Teams. As a result, you will need to bring a phone, tablet or laptop with you to class to participate in the discussion. Further, as described below, you will need to use your phone, tablet or laptop to register your in-person attendance via iClicker Cloud to earn the participation point for the session.

Please do ask the guest speakers and / or me questions via the Piazza Live Q&A. Remember, if you are wondering about something, chances are someone else is too! Additionally, this provides a backup to the iClicker Cloud for attendance.

Learning Resources, Technology & Texts

There are no texts for the senior seminar course.

We will make use of Brightspace (https://purdue.brightspace.com), look for Spring 2026 AAE 40000-001 LEC; Piazza (see class link from Brightspace); and iClicker Cloud (https://student.iclicker.com/).

Learning Outcomes

By the end of the seminar course, you will be able to:

1. Identify multiple career paths pursued by people with aerospace engineering degrees; including those of people still working in aerospace and those of people who used their aerospace engineering degree as a basis for a career in a different field
2. Recognize the broad range of products, topics, applications and systems on which aerospace engineers work
3. Evaluate issues related to engineering ethics using professional society codes
4. Have an awareness of issues related to intercultural teams

Earning a Grade in the Senior Seminar

The senior seminar course has one credit associated with it, so you will earn a letter grade. **The course relies entirely upon attendance and / or participation; there are no typical assignments.**

You have three ways of earning points: (1) attending the seminar class meetings with a speaker or participating in the activity for weeks that do not have a speaker; (2) writing an essay; or (3) attending an in-person or a virtual colloquium, seminar or similar talk whose topic relates to engineering. Details of these appear below.

Participation Points: Class Attendance Option (9 points minimum)

There are 15 meetings of the seminar course scheduled this semester; attending one class session or completing the corresponding activity earns you one participation point. To record your

attendance at class sessions with an in-class component, we will use iClicker Cloud, which uses geolocation to verify that you are in the classroom for the session. You can do this with the same phone, tablet or laptop that you will bring to participate in the discussions via Piazza Live Q&A. Reiterating the point above, you will need to bring a phone, tablet or laptop with you to class.

If you miss class for any reason, including medically excused absences, you can use the "write an essay" or "attend a talk" options below to earn additional participation points. You can earn a maximum of five points total via any combination of the "write an essay" or the "attend a talk" options.

You must earn a minimum of 9 points by participating in-person; i.e., if you attend nine or fewer classes, you will fail, regardless of other options below. You are responsible for ensuring that your attendance is properly recorded.

Participation Points: Attend a Talk

You can also earn participation points by attending a seminar, colloquium or other type of talk related to engineering. If you use this option, you need to submit a document on Brightspace via the assignment at: Content > Other ways of earning participation points > Option: Attend a talk. This cannot be a part of another course for which you are registered; the seminar course tries to broaden your perspective about engineering in general and aerospace engineering in particular.

In your document, you need to respond to all the prompts indicated on Brightspace. In addition to the response to the prompts, you must submit a photo or selfie of yourself at the in-person event, or a screen capture of your participation in a remote or web-based event (selfies of you in front of your monitor during the talk will work, too), and you must provide the date of the class session you are replacing. You will not receive credit if any of the required items are missing.

You must submit your document within two weeks of the talk you attended, or it will not count. I will not accept these after 11:59pm Eastern Tue 05 May 2026.

Participation Points: Essay Option

Using this option, you can compensate for missed classes by writing a 1500-word essay. There is a Brightspace assignment at "Content > Other ways of earning participation points > Option: Write an essay" that lets you suggest topics and then submit your essay.

Step 1: Suggest three different topics that you think are relevant to AAE 400. The essay topics can be related to any aerospace engineering-related or more general engineering-related topic in which you are interested. When you submit topics using step 1 of the assignment, add a short paragraph describing the scope of each topic.

Step 2: Once your topic is approved, you must upload your essay on Brightspace using step 2 of the assignment.

The essay is due either: (a) two weeks after I provide you feedback and / or recommend a topic from your suggestions, or (b) by the last acceptance day shown below, whichever is soonest.

You must include the word count and the date of the session you wish to replace at the top of your post. Your essay is to be 1500 words long, with a few words margin above and below this target. Your essay must be properly formatted, present sound reasoning, and be essentially free of spelling and grammatical errors. The work must be demonstrably your own. This cannot be work turned in for a grade in another course. You are not to use AI to write the essay for you. I will use Turnitin via Brightspace to check for plagiarism and to provide a likely generated by AI score. I recognize the challenges with AI checking software, but students have previously simply copied output from ChatGPT as an essay. This is not demonstrably your own work.

I will accept the list of three topics no later than 11:59pm Eastern Tue 21 Apr 2026, and I will accept the essays no later than 11:59pm Eastern Tue 05 May 2026.

Grading Scale

Because of the participation point-based approach to the grade, I will use the following straight-scale grading scheme. There will be no "+" or "–" grades, except for the A+, which you can earn by doing something more than attending every class session.

Participation Points	Letter Grade
15	A+
14	A
12 or 13	B
10 or 11	C
9	D
8 or below	F

I will adjust the scale here if unforeseen circumstances lead to less than 15 equivalent class sessions.

Attendance Policy

This course follows the University Academic Regulations regarding class attendance, which state that students are expected to be present for every meeting of the classes in which they are enrolled. Attendance is recorded via iClicker Cloud as described above. As also described above, you may miss some course sessions and replace those points with other options, if you have attended at least nine sessions in person.

The options for "other ways to earn points" also cover absences that are excused (grief/bereavement, military service, jury duty, parenting leave, or emergent medical care) as described at the Office of the Dean of Students (ODOS) website.

Course Schedule

Most class meetings will have a guest speaker or some other in-class opportunity; one week will have an activity that you will complete on your own and that week will not have an in-person meeting. Guest speakers may have unforeseen issues that preclude their participation with short notice or may require them to participate remotely. I will adjust the topics and speakers as necessary. Please see the Brightspace site for additional details as they become available.